



Zahran Toric Tool

USER MANUAL · V1.0.0 · IOS &
ANDROID

Clinical Decision Support Tool
For ophthalmic surgeons



COMPLETE CLINICAL WORKFLOW

- Create Patient** — Name, DOB, notes. Tap + *New Patient* from Patient List.
- Add Eye Record** — Enter K1/K2 powers & axes, IOL power, IOL cylinder, target axis, SIA.
- Run Toric Calculator** — Enable PCA toggle if no posterior corneal data. Review target axis & residual astigmatism.
- Capture Eye Photo** — Align limbus to gold ring, pupil to cyan ring. Wait for **LEVEL** in centre, then tap shutter.
- Annotate Landmarks** — Tap vessels on photo to mark reference points. Tap Save. (Compensates for intraoperative cyclorotation.)
- Align IOL** — Use Alignment Screen overlay in OR. Match IOL marks to green target axis line.
- Post-op Analysis** — Enter post-op refraction in Alpins screen. Review CI, ME, AE, IS metrics.
- Log SIA Case** — Add case to SIA Nomogram to build personal surgeon profile over time.



SCREEN NAVIGATION

SCREEN	ACCESS FROM
Patient List	App home screen
Patient Form	+ New / Edit button
Patient Detail	Tap any patient
Toric Calculator	Patient Detail → Calc
Camera	Patient Detail → Camera
Alignment	After photo capture
Alpins Analysis	Patient Detail → Alpins
Landmark Annotation	Patient Detail → Annotate
SIA Nomogram	Patient List → SIA button
Settings	Patient List → ⚙ icon



EYE RECORD FIELDS

FIELD	UNIT
Eye side	OD / OS
K1 Power (flat)	Dioptres
K1 Axis	° TABO
K2 Power (steep)	Dioptres
IOL Power	Dioptres
IOL Cylinder	Dioptres
Target Axis	° TABO
SIA / SIA Axis	D / °
Post-Refractive	Toggle + LASIK/PRK/RK

△ K1 = *flat* meridian (lower D). K2 = *steep* (higher D).



TORIC CALCULATOR

- Pre-populates from eye record automatically
- PCA Toggle** — adds Barrett posterior corneal ATR contribution ($\approx +0.3$ D) using double-angle vectors. Enable unless you have Pentacam TCRP data.
- Post-Refractive banner** — shown when eye had prior LASIK/PRK/RK. Use Barrett True-K or Haigis-L instead.
- Rotation Effect Chart** — residual vs. IOL rotation 0° – 30° . Steep slope = sensitivity to rotation.
- Export Plan** — generates PDF with biometry, axes, rotation chart. Share via email/AirDrop.

```
// PCA double-angle correction
Tx = Ax + 0.3 | Ty = Ay
mag = sqrt(Tx^2+Ty^2)
axis = 0.5*atan2(Ty,Tx)*180/pi [mod 180]
```



CAMERA — INSTRUMENT GAUGE

ELEMENT	MEANING
Gold triple rings	Decorative outer border
Main gold ring	Align limbus to this
Gold tick marks	Every 5° (major at 30°)
TABO labels	0° – 180° around ring
Cyan dashed ring	Align pupil margin here
Centre readout	Live tilt $^{\circ}$, turns green $\leq \pm 3^{\circ}$
Green shutter	Device is level — capture now



ALIGNMENT SCREEN OVERLAY

COLOUR	LINE / ELEMENT
Blue	Reference (K1) axis
Green	Target IOL axis — align IOL markings here
Gold	Current adjusted axis
Cyan ring	Limbal registration guide
Coloured dots	Saved vessel landmarks

- Use **+/-** buttons to rotate overlay in 1° steps
- Recapture** — new photo without losing record
- Annotate** — open landmark annotation



LANDMARK ANNOTATION

- Tap on the eye photo to place a marker (ring + dot + number)
- Colours cycle: red → gold → green → blue → pink → orange
- Undo Last** — removes most recent marker
- Clear All** — removes all markers
- Tap **Save** — markers persist on alignment overlay

Mark 3–4 distinct conjunctival vessels at different clock positions (e.g. 3h, 6h, 9h). Avoid marking the limbus itself.



ALPINS VECTOR ANALYSIS

METRIC	IDEAL	GREEN	YELLOW	RED
TIA Target-Induced Astigmatism	—	Planned correction (from K data)		
SIA Surgically-Induced Astigmatism	—	Actual change (from post-op Rx)		
DV Difference Vector (residual)	0.00 D	TIA tip → SIA tip distance		
CI Correction Index	1.00	0.9–1.1	0.75–1.25	<0.75 or >1.25
ME Magnitude Error	0.00 D	≤±0.25 D	>±0.25 D	—
AE Angle Error	0.0°	≤±5°	>±5°	—
IS Index of Success	0.00	<0.2	0.2–0.5	>0.5

CI Interpretation

>1.0	Overcorrected	IOL cylinder too strong or over-rotated
= 1.0	Perfect	Exact target achieved
<1.0	Undercorrected	Consider higher cylinder next case

Double-Angle Plot

- **Blue dashed** — TIA vector
- **Green dashed** — SIA vector
- **Red solid** — DV (residual error)
- Short DV = good outcome. Long DV = residual error.

Enter post-op refraction cylinder as SIA for real analysis. Pre-filled IOL cylinder = theoretical only.



SIA NOMOGRAM TRA

- Access via Patient List → SIA
- Log each case: planned SIA magnitude/axis + achieved SIA magnitude/axis
- App calculates **vector mean** cases
- Case accuracy colour coding: D · ● ≤0.50 D · ● >0.50 D

Build ≥20 cases before using mean routine calculations. Consider separate profiles for different incision sites.

SETTINGS

SETTING	DESCRIPTION
Default SIA	Pre-fills SIA field in new records
Default SIA Axis	Pre-fills SIA axis (your usual incision site)
PCA Correction	Global default for PCA toggle
Surgeon Name	Appears on exported PDF plans
Institution	Appears on exported PDF plans



TABO AXIS REFERENCE

DEGREES	CLOCK	DIRECTION
0° / 180°	3h / 9h	Horizontal
45°	1:30h	Oblique up-right
90°	12h	Vertical (superior)
135°	10:30h	Oblique up-left

- **WTR** — steep meridian near 90° (vertical)
- **ATR** — steep meridian near 0°/180° (horizontal)
- PCA adds ≈0.3 D ATR — reduces effective WTR, increases ATR

TROUBLESHOOTING

PROBLEM	CAUSE	FIX
Camera screen black	Camera permission denied	Settings → Privacy → Camera → enable for Zahran Toric Tool
Tilt not responding	Motion sensor unavailable	Restart app; ensure motion access is granted
Calculator result incorrect	K1 entered as steeper meridian	K1 = flat (lower D); K2 = steep (higher D). Swap if needed.
Alpins CI very high (>2)	SIA uses IOL cylinder default	Enter actual post-op refraction cylinder for real analysis
Landmarks missing on alignment	Not saved before exiting	Re-open Landmark Annotation and tap Save
PCA toggle not visible	K values missing from eye record	Fill K1 Power, K1 Axis, and K2 Power in the eye record first
PDF export fails	No share target installed	Restart app; ensure Files or email app is available



PERMISSIONS & DATA

PERMISSION	USE	STATUS
Camera	Capture eye photos	Required
Photo Library	Save to gallery (optional)	Optional
Motion Sensors	Level indicator	Required

- All data stored **locally on device**
- No internet connection required
- Export PDFs before uninstall — uninstall deletes all data